

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250277992

Kind Code

A1

Publication Date

September 04, 2025

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EYEWEAR ASSEMBLY

Abstract

An eyewear assembly including primary lenses in the form of eyeglasses, secondary lenses engageable with the eyeglasses, and a connector assembly. The secondary lenses may be sunglasses or any of a range of therapeutic lenses other than sunglasses. The connector assembly includes a magnetic member and a non-magnetic member. The magnetic member is provided in one of the primary and secondary lenses, and the non-magnetic member is provided in the other of the primary and secondary lenses. Each magnetic member attracts the associated non-magnetic member to itself when the secondary lenses are brought into close enough proximity to the primary lenses. The magnetic and non-magnetic members thereby detachably retain the primary and secondary lenses in engagement with one another.

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Family ID: 96881121

Appl. No.: 18/594131

Filed: March 04, 2024

Publication Classification

Int. Cl.: G02C9/04 (20060101); G02C7/08 (20060101); G02C7/10 (20060101); G02C13/00 (20060101)

U.S. Cl.:

CPC G02C9/04 (20130101); G02C7/086 (20130101); G02C7/10 (20130101); G02C13/001 (20130101); G02C2200/02 (20130101)

Background/Summary

TECHNICAL FIELD

[0001] The present disclosure is directed to eyewear. More particularly, the present disclosure relates to an eyewear assembly which includes primary lenses and secondary lenses. Specifically, the disclosure is directed to an eyewear assembly which includes primary lenses in the form of eyeglasses, a secondary lenses which are engageable with the eyeglasses, and a connector assembly. The secondary lenses may be sunglass lenses or any one of a range of therapeutic lenses for concussion, migraine etc. that are a specified tint not used for sunglasses. Others secondary lenses could include a filter for blue light emitted by monitors, clear, sunglass bifocals and readers for presbyopic people. The connector assembly includes a magnetic member and a non-magnetic member. The magnetic member is provided in one of the primary and secondary lenses, and the non-magnetic member is provided in the other of the primary and secondary lenses. The magnet attracts the non-magnetic member to itself when the secondary lenses are brought into close enough proximity to the primary lenses. The magnetic and non-magnetic members detachably retain the primary and secondary lenses in engagement with one another.

BACKGROUND ART

[0002] It is known in the art to provide sunglasses with tinted prescription lenses which are able to be used in bright light conditions instead of regular prescription eyeglasses. It is further known in the art to provide a substitute for tinted prescription lens sunglasses by providing non-prescription lens sunglasses which are selectively engaged in some fashion with prescription eyeglasses. The sunglasses may, for example, be of type that can be clipped onto a frame of the eyeglasses to form an eyewear assembly that can then be used in bright light conditions. The eyewear assembly allows users to wear their prescription eyeglasses but offers a lower cost option than would be the case if the person had to purchase separate prescription sunglasses. Detachable sunglasses also enable users to own multiple pairs of different style sunglasses simply by replacing one clip-on set with another. Stock clip-on sunglasses are readily available but typically do not fit existing frames very well. Furthermore, they tend to come in limited colors of lenses and chassis. Custom-made clip-on sunglasses are readily available at higher cost than stock clip-on sunglasses. These more expensive clip-ons are customized in a variety of ways and are attachable to existing prescription glasses in a variety of ways. Most of these attachment systems require special components to be made to order by the supplier. They tend to be difficult for an optician or optometrist to assemble and many are not durable.

[0003] Other methods of attaching sunglasses to eyeglasses are known including providing magnets on each of the sunglasses and the eyeglasses. The sunglasses are then retained on the eyeglasses through magnetic attraction between the magnets. One of the issues with such systems comes from the nature of magnets themselves. Magnets all have a north pole, a south pole, and they exude a magnetic field. The sunglasses and eyeglasses will only be able to become engaged with one another if the magnets in the sunglasses and eyeglasses are oriented in a very specific fashion. The magnets have to be arranged so that their opposite poles are brought into close proximity to one another. In other words, the north pole of one magnet must be able to be brought into close proximity with the south pole of the other magnet. If, however, the magnets are arranged so that their like poles will be brought into close proximity to one another, then the eyeglasses and sunglasses will repel one another and will not be able to become engaged with one another. Correctly orienting the magnets in the eyeglasses and sunglasses is therefore critical.

SUMMARY OF THE INVENTION

[0004] The present disclosure relates to an eyewear assembly which addresses some of the problems of prior art eyewear assemblies. An eyewear assembly including primary lenses in the form of eyeglasses, secondary lenses engageable with the eyeglasses, and a connector assembly. The secondary lenses may be sunglasses or any of a range of therapeutic lenses for concussion, migraine etc. that are a specified tint not used for sunglasses. Others secondary lenses could include

a filter for blue light emitted by monitors or be clear, sunglass bifocals and readers for presbyopic people. The connector assembly includes a magnetic member and a non-magnetic member. The magnetic member is provided in one of the primary and secondary lenses, and the non-magnetic member is provided in the other of the primary and secondary lenses. The magnet attracts the non-magnetic member to itself when the secondary lenses are brought into close enough proximity to the primary lenses and thereby detachably retain the primary and secondary lenses in engagement with one another.

[0005] The non-magnetic member is fabricated preferably from a metal. In one embodiment, the non-magnetic member may be stainless steel. In another embodiment, the non-magnetic member may be a metal other than stainless steel. In a further embodiment, the non-magnetic member may be a coated metal. It will be understood that any suitable non-magnetic member may be used that will, with the magnetic member, hold the primary and secondary lenses in abutting contact with one another. Because the eyewear assembly uses only one magnet, the orientation of the magnet used in the one of the primary lenses and the secondary lenses is not critical. If the magnet presents a north pole or a south pole to the non-magnetic member, the non-magnetic member will still be coupled to the magnetic member and thus the secondary lenses will still be retained in engagement with the primary lenses. It is furthermore not critical if the magnet is installed in the primary lenses or the secondary lenses as either installation will function if the non-magnetic member is installed in the other of the primary lenses or the secondary lenses.

[0006] The magnetic member or non-magnetic member is received in a slot defined in a perimeter of the lenses of the eyeglasses and is retained therein by cooperative engagement of the eyeglass frame with the lens. The other of the magnetic member and the non-magnetic member is received in an aperture defined in the associated secondary lens. The aperture is created in a location which will align with the slot when the secondary lenses are brought into close proximity with the primary lenses. The magnet attracts the non-magnetic member to itself when the secondary lenses are brought into close enough proximity to the primary lenses and thereby detachably retains the primary and secondary lenses in engagement with one another.

[0007] The primary lenses as described herein may be referred to herein as “eyeglasses” and the secondary lenses as described herein may be referred to as “sunglasses”. It should, however, be understood that the two terms are not to be narrowly construed as eyeglasses with corrective or prescription lenses and sunglasses with lenses tinted to block Ultra Violet light. Rather, the terms should be understood as two lens types having different properties from one another that are contemplated as being useful to couple together to aid a person in some manner. In one embodiment, the eyeglasses include primary lenses which are prescription lenses and the secondary lenses are sunglasses but could be any one of a range of therapeutic lenses for concussion, migraine etc. that are a specified tint not used for sunglasses. Others secondary lenses could include a filter for blue light emitted by monitors, clear, sunglass bifocals and readers for presbyopic people.

[0008] In one aspect, an exemplary embodiment of the present disclosure may provide an eyewear assembly comprising primary lenses, secondary lenses, and a connector assembly detachably securing the primary lenses and the secondary lenses to one another; wherein the connector assembly comprises a magnetic member provided in a lens of one of the primary lenses and the secondary lenses; and a non-magnetic member provided in a lens of the other of the primary lenses and the secondary lenses.

[0009] In one embodiment, the non-magnetic member may be comprised of a material which is attractable to the magnetic member. In one embodiment, the material may be a metal. In one embodiment, the metal may be stainless steel. In one embodiment, a region of the magnetic member may extend outwardly beyond a surface of an associated lens of the one of the primary lenses and the secondary lenses. In one embodiment, a region of the non-magnetic member may extend outwardly beyond a surface of an associated lens of the other of the primary lenses and the secondary lenses. In one embodiment, an end surface of the magnetic member in the one of the

primary lenses and the secondary lenses may contact an end surface of the non-magnetic member in the other of the primary lenses and the secondary lenses when the secondary lenses are operatively engaged with the primary lenses. In one embodiment, the eyewear assembly may further comprise a space defined between an outer surface of the lens of the primary lenses and an inner surface of the lens of the secondary lenses.

[0010] In one embodiment, a notch may originate in a peripheral edge of the lens of the primary lenses and the one of the magnetic member and the non-magnetic member is received in the notch. In one embodiment, the notch may extend between an outer surface and an inner surface of the lens of the primary lenses. In one embodiment, a region of a wall of the lens which defines the notch may be complementary in configuration to a portion of a circumferential surface of the one of the magnetic member and the non-magnetic member received in the notch. In one embodiment, the eyeglass assembly may further comprise a securement mechanism provided on the primary lenses, wherein the securement mechanism retains the one of the magnetic member and the non-magnetic member in the notch. In one embodiment, the securement mechanism may comprise a portion of a frame of the primary lenses. In one embodiment, the securement mechanism may comprise a length of a nylon wire provided on the primary lenses. In one embodiment, an aperture may extend between an outer surface and an inner surface of the lens of the secondary lenses, wherein the aperture is located a distance inwardly away from a peripheral edge of the lens of the secondary lenses, and wherein the other of the magnetic member and the non-magnetic member is received in the aperture.

[0011] In one aspect, an exemplary embodiment of the present disclosure may provide a method comprising providing primary lenses including at least one eyeglass lens; providing secondary lenses including at least one sunglass lens; providing a magnetic member; providing a non-magnetic member; installing one of the magnetic member and the non-magnetic member on the at least one eyeglass lens; installing the other of the magnetic member and the non-magnetic member on the at least one sunglass lens; attracting the non-magnetic member to the magnetic member; and retaining the secondary lenses to the primary lenses.

[0012] In one embodiment, the method further comprises defining a notch in the at least one eyeglass lens; defining an aperture in the at least one sunglass lens; installing the one of the magnetic member and the non-magnetic member in the notch defined in the at least one eyeglass lens; and installing the other of the magnetic member and the non-magnetic member in the aperture defined in the at least one sunglass lens. In one embodiment, the method may further comprise moving the secondary lenses and primary lenses into close proximity with one another and aligning the magnetic member in the one of the primary lenses and the secondary lenses with the non-magnetic member in the other of the primary lenses and the secondary lenses. In one embodiment, the method may further comprise bringing the magnetic member and the non-magnetic member into direct contact with one another. In one embodiment, providing the non-magnetic member may include providing a rod of stainless steel capable of being attracted to a magnet.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0013] Sample embodiments of the present disclosure are set forth in the following description, are shown in the drawings and are particularly and distinctly pointed out and set forth in the appended claims.

[0014] FIG. 1 is a front, top, right side perspective view of a first embodiment of eyeglasses in accordance with an aspect of the present disclosure;

[0015] FIG. 2A is a partially exploded front, top, right side perspective view of the eyeglasses of FIG. 1;

[0016] FIG. 2B is a partially exploded front, top, right side perspective view of a pair of lenses for the eyeglasses, shown in isolation;

[0017] FIG. 3 is a front elevation view of the eyeglasses shown in isolation;

[0018] FIG. 4A is a cross-section of a region of the eyeglasses taken along line 4A-4A of FIG. 3;

[0019] FIG. 4B is a cross-section of a region of the eyeglasses taken along line 4B-4B of FIG. 3;

[0020] FIG. 5 is an enlarged, exploded front elevation view of a first embodiment of a single lens of the eyeglasses shown in FIG. 3;

[0021] FIG. 5A is a partial enlarged, exploded front elevation view of a second embodiment of a single lens of the eyeglasses shown in FIG. 3;

[0022] FIG. 5B is a partial enlarged, exploded front elevation view of a third embodiment of a single lens of the eyeglasses shown in FIG. 3;

[0023] FIG. 5C is a partial enlarged, exploded front elevation view of a fourth embodiment of a single lens of the eyeglasses shown in FIG. 3;

[0024] FIG. 6 is a front, top, right side elevation view of the eyeglasses of FIG. 3 showing a first embodiment of sunglasses in accordance with an aspect of the present disclosure exploded outwardly away therefrom;

[0025] FIG. 7 is a front, top, right side elevation view of the eyeglasses of FIG. 3 showing the first embodiment of the sunglasses of FIG. 6 operatively engaged therewith;

[0026] FIG. 8A is a cross-section of the eyeglasses and the first embodiment sunglasses taken along line 8A-8A of FIG. 6;

[0027] FIG. 8B is a cross-section of the eyeglasses and the first embodiment sunglasses taken along line 8B-8B of FIG. 7;

[0028] FIG. 9 is a front, top, right side elevation view of the eyeglasses of FIG. 3 showing a second embodiment of sunglasses in accordance with an aspect of the present disclosure exploded outwardly away therefrom;

[0029] FIG. 10 is a cross-section of the eyeglasses and the first embodiment sunglasses similar to FIG. 8B except showing the steel rod located in the eyeglasses and the magnet located in the first embodiment sunglasses;

[0030] FIG. 11 is a front elevation view of a second embodiment of eyeglasses in accordance with an aspect of the present disclosure;

[0031] FIG. 11A is a cross-section through a lens of the eyeglasses taken along line 11A-11A of FIG. 11; and

[0032] FIG. 11B is a cross-section through a lens of the eyeglasses taken along line 11B-11B of FIG. 11.

[0033] Similar numbers refer to similar parts throughout the drawings.

DETAILED DESCRIPTION

[0034] The present description relates to an eyewear assembly comprising primary lenses, secondary lenses, and a connector assembly for operably detachably engaging the primary lenses and secondary lenses to one another. For ease of description, the primary lenses will be referred to herein as “eyeglasses” and the secondary lenses will be referred to herein as “sunglasses”. It will be understood that any suitable types of lenses for improving or enhancing the vision of a user are contemplated to be usable as either of the primary lenses or the secondary lenses.

[0035] As will be described hereafter, the sunglasses are retained in operable engagement with the eyeglasses utilizing the connector assembly. The connector assembly includes at least one magnet provided in each lens of the eyeglasses or the sunglasses, and a non-magnet provided in each lens of the other of the eyeglasses and the sunglasses. The non-magnet may be in the form of a rod fabricated from a metal that is able to be attracted to a magnet. The attraction between the magnet(s) and non-magnet(s) holds the primary lenses and secondary lenses adjacent one another. These various components and how they interact with one another will be described hereafter. It will be understood that the magnet(s) may be provided on the eyeglasses or the sunglasses and that

the non-magnetic rods will then be provided on the other of the eyeglasses and the sunglasses. The following describes an embodiment where the magnets are provided on the eyeglasses and the non-magnetic rods are provided on the sunglasses.

[0036] Referring to FIGS. 1-5, there is shown a first embodiment of eyeglasses in accordance with an aspect of the present disclosure, generally indicated at **10**. Eyeglasses **10** comprise a frame **12**, a pair of arms **14** which are pivotally engaged with opposite sides of the frame **12**, a first lens **16**, and a second lens **18**. First lens **16** and second lens **18** may be prescription lenses.

[0037] Frame **12** comprises a first region **12a** having a front surface **12a'**, a rear surface **12a''**, and the first region **12a** defines a first aperture **12a''** therein. Similarly, frame **12** includes a second region **12b** which has a front surface **12b'** and a rear surface (not shown), and the second region **12b** defines a second aperture **12b''** therein. A bridge **12c** extends between first region **12a** and second region **12b**. When eyeglasses **10** are worn, bridge **12c** will rest upon a person's nose and the two arms **14** will extend along opposing sides of the person's head and rest on a top of an associated ear. It will be understood that the configuration of frame **12** illustrated in the attached figures is exemplary only and that any desired configuration of frame **12** may be utilized in the performance of the disclosed eyewear assembly and method.

[0038] First lens **16** and second lens **18** are each configured so as to be complementary to the shape and size of the respective first aperture **12a''** and second aperture **12b''** defined by frame **12**. In accordance with a specific aspect of the present disclosure and as best seen in FIGS. 2B-4B, each of the first lens **16** and second lens **18** has an outer surface **16a**, **18a** and an inner surface **16b**, **18b** (FIGS. 2 and 4A). A beveled peripheral edge **16c** (FIG. 2A) extends between outer surface **16a** and inner surface **16b** of the first lens **16**. Similarly, a peripheral edge **18c** (FIG. 2A) extends between outer surface **18a** and inner surface **18b** of second lens **18**.

[0039] As best seen in FIG. 4A, the peripheral edge **16c** (and similarly the peripheral edge **18c**) is configured to be complementary in profile to the wall of frame **12** which defines first aperture **12a''** (and similarly second aperture **12b''**). FIG. 4A shows that the wall of first region **12a** of frame **12** which defines first aperture **12a''** is configured to define a V-shaped depression **12d** therein. The wall defining the V-shaped depression **12d** therein circumscribes the first aperture **12a''**. The wall of frame **12** which defines second aperture **12b''** is similarly configured. The profiles of the peripheral edges **16c**, **18c** of the first lens **16** and second lens **18** are complementary to the profiles of the walls defining first aperture **12a''** and second aperture **12b''**. When first lens **16** is received within first aperture **12a''** of frame **12**, the V-shaped profile of the beveled peripheral edge **16c** of first lens **16** is received in the complementary V-shaped depression **12d** defined in the wall which circumscribes first aperture **12a''**. This complementary matching configuration helps retain first lens **16** in engagement with frame **12**. Second lens **18** is similarly retained in engagement with frame **12**.

[0040] It will be understood that frame **12** may define any shape and size opening **12a**, **12b** therein and that the wall of frame **12** which defines those openings **12a**, **12b** may have any profile that is desired in order to keep first lens **16** and second lens **18** engaged therewith. The profile of the wall defining the apertures **12a''**, **12b''** may therefore be formed with a depression having a different configuration than the V-shape illustrated in the attached drawings. For example, the depression formed in the wall bounding the apertures **12a''**, **12b''** may be square, or U-shaped. The peripheral edge **16c**, **18c** of the first and second lenses **16**, **18** will then be a complementary square-shape or U-shaped. Alternatively, the depression may be formed on the peripheral edge **16c**, **18c** and a complementary projection may be provided on the wall which bounds and defines the apertures **12a''**, **12b''**. The first and second lenses **16**, **18**, for example, may be formed with a V-shaped, square, or U-shaped depression in their peripheral edges **16c**, **18c** and the wall of frame **12** which defines the respective apertures **12a''**, **12b''** may be formed with a projection which fits complementarily into the depression defined in the lenses **16**, **18**.

[0041] In accordance with an aspect of the present disclosure, a notch **16d** (FIG. 2B) is formed in

the peripheral edge **16c** of first lens **16** and extends for a distance inwardly into the body of the first lens **16**. notch **16d** extends between outer surface **16a** and inner surface **16b** of first lens **16**. notch **16d** preferably is formed on a section of peripheral edge **16c** of first lens **16** which will be engaged with a side of the first region **12a** of frame **12** that is located remote from bridge **12c**. Similarly, a notch **18d** is formed in the peripheral edge **18c** of second lens **18** and extends for a distance inwardly into the body of the second lens **18**. notch **18d** extends between outer surface **18a** and inner surface **18b** of second lens **18**. notch **18d** preferably is formed on a section of peripheral edge **18c** of second lens **18** which will be engaged with a side of the second region **12b** of frame **12** remote from bridge **12c**. Preferably, first notch **16d** and second notch **18d** are laterally aligned with one another so that the eyeglasses have a symmetrical in appearance relative a vertical axis passing through the bridge **12c**. This symmetrical arrangement will give eyeglasses **10** a more aesthetically pleasing appearance.

[0042] It will be understood that in other embodiments, notches **16d**, **18d** may be formed at any other desired location on the peripheral edge **16c**, **18d** of the associated one of first lens **16** and second lens **18**. It will further be understood that more than one notch **16d**, **18d** may be formed on the respective peripheral edge **16c**, **18d** of the associated one of first lens **16** and second lens **18**. If more than one notch **16d**, **18d** is defined in the associated lenses **16**, **18**, it will be understood that any notch(s) may be defined in the side of the peripheral edge **16c**, **18c** remote from bridge **12c** and/or proximate bridge **12c**, and/or either along a top edge and/or bottom edge of lenses **16**, **18**. It will be further understood that in other embodiments, the location of the notch(s) **16d** in first lens **16** may not be mirrored in the location of notch **18d** in second lens **18**. In other words, first lens **16** and second lens **18** may not be symmetrical relative to a vertical axis passing through bridge **12c**.

[0043] As best seen in FIG. 2B, in the first embodiment eyeglasses **10**, each notch **16d**, **18d** is generally U-shaped when first lens **16** and second lens **18** are viewed from a direction looking at the outer surfaces **16a**, **18a** thereof. The U-shape originates in the peripheral edge **16c**, **18c** and extends for a distance inwardly into the body of the associated lens **16**, **18**. The curved surface of the U-shape is located a distance inwardly away from the peripheral edge **16c**, **16d**.

[0044] It will be understood that the U-shape of the notches **16d**, **18d** is only exemplary of one of many suitable shapes in which the notches may be formed. In other embodiments, the shape of each notch **16d**, **18d** may be other than U-shaped. A small sample of other exemplary shaped notches are illustrated in FIGS. 5A, 5B, and 5C. FIG. 5A, for example, shows a second embodiment lens **116** defining a square-shaped notch **116d** therein. FIG. 5B shows a third embodiment lens **216** defining a U-shaped notch **216d** wherein the innermost edge of the notch is comprised of three-straight sections instead of a curve. FIG. 5C shows a fourth embodiment lens **316** defining an U-shaped notch **316d** which is wider than notch **16d** and is tending to be generally elliptical in nature. Although not illustrated herein, it will be understood that the notches may, alternatively, be keyhole shaped, triangular, rectangular, or even irregular in shape. It will be understood that any desired shape of notch may be formed in first lens **16** and second lens.

[0045] In accordance with one aspect of the present disclosure a first magnet **20** is received in the notch **16d** of first lens **16** and a second magnet **22** is received in the notch **18d** of second lens **18**. Magnets **20**, **22** are shaped and sized such that they are received in the associated notches **16d**, **18d** and are retained therein by a portion of the frame **12** of eyeglasses **10**. The portion of the frame **12** acts as a securement mechanism which keeps the magnet **20** or magnet **22** within the associated lens **16**, **18**. If the frame **12** and one or both lenses **16**, **18** are disengaged from one another, the associated magnet **20** or magnet **22** will tend to fall out or drop out of the associated notch **16d**, **18d**. FIG. 2B shows that each of the first and second magnets **20**, **22** is generally cylindrical in shape. Utilizing a generally U-shaped notch **16d**, **18d** and cylindrical magnet with a circular cross section tends to reduce stress in the lenses **16**, **18** but any configuration notch and complementary cross-section of magnet is contemplated for the disclosed eyewear assembly and method.

[0046] FIGS. 2B and 4B show that first magnet **20** has a first surface **20a** and an opposed second

surface **20b** with a circumferential wall **20c** extending between first surface **20a** and second surface **20b**. Similarly, second magnet **22** has a first surface **22a** and an opposed second surface **22b** with a circumferential wall **22c** extending between first surface **22a** and second surface **22b**. Circumferential walls **20c**, **22c** have a radius of curvature which is complementary to the curved innermost edge of the associated notch **16d**, **18d**. FIG. 5, for example, shows the curved innermost edge **16d'** of the wall of lens **16** which defines notch **16d**. The wall of second lens **18** which defines notch **18d** will have a substantially similar curved innermost edge having a radius of curvature which is complementary to the radius of curvature of the circumferential wall **22c** of second magnet **22**.

[0047] In order to install first magnet **20** within notch **16d**, first magnet **20** is slid into the opening to notch **16d** in the direction indicated by arrow "A" in FIG. 5. The diameter of first magnet **20** is such that the magnet may be at least partially retained within notch **16d** by friction. When the lenses **16**, **18** are installed within frame **12**, as described later herein, the frame **12** keeps each magnet **20**, **22** retained in the associated lens **16**, **18**.

[0048] It should be noted that while every magnet **20**, **22** has a north pole and a south pole, the orientation of the poles in the installation of the magnets **20**, **22** is completely irrelevant since the components which are to interact with the magnets **20**, **22** in the present disclosure are not, themselves, magnetic but are instead non-magnetic.

[0049] FIGS. 5A, 5B and 5C show different configurations of first magnets, namely magnets **120**, **220**, and **320** being inserted into substantially complementary notches **116d**, **216d**, and **316d** in the second, third, and fourth embodiments of first lenses **116**, **216**, and **316**, respectively. The magnets **120**, **220**, **320** can be slid into an out of the associated notches **116d**, **216d**, **316d** as indicated by the arrows "A". The notch **116d** in lens **116** is substantially square in configuration and the first magnet **120** which is receivable in notch **116d** is substantially square in configuration. The notch **216d** in lens **216** is substantially U-shaped with an innermost edge of the wall defining the notch **216d** having three straight sections of substantially equal length. The first magnet **220** which is receivable in notch **216d** is substantially hexagonal in configuration. When first magnet **220** is received in notch **216d** three of the six faces nesting with the innermost edge of the wall defining notch **216d**. The notch **316d** in lens **316** is a wide U-shaped notch and the magnet **320** that is configured to be received in the notch **316d** is generally elliptical in shape. The magnet **320** has a curved side edge which will nest with the innermost edge of the wall defining the notch **316d** when magnet **320** is received therein.

[0050] FIG. 4A shows the first magnet **20** of the first embodiment eyeglasses **10** received within the notch **16d** defined by first lens **16**. As is evident FIG. 4A, first region **12a** (and second region **12b**) of frame **12** is of a thickness "T1" measured between the front surface **12a'** and rear surface **12a''** thereof. First lens **16** is of a thickness "T2" measured from outer surface **16a** to inner surface **16b** thereof. The thickness "T1" is greater than the thickness "T2". FIG. 4B shows that first magnet **20** is of a thickness "T3" as measured from first surface **20a** to second surface **20b** thereof. The thickness "T3" is greater than the thickness "T2" and may be equal to or greater than the thickness "T1". In other words, magnet **20** (and magnet **22**) is of a greater thickness than the first lens **16** (and second lens **18**) and may be of a thickness that is equal to or greater than that of the frame **12**. The second surface **20b** of magnet **20** is substantially flush with the inner surface **16b** of lens **16** and the first surface **20a** of magnet **20** projects a distance "D" outwardly beyond the outer surface **16a** of lens **16**. In some embodiments, the first surface **20a** of magnet **20** is flush with the front surface **12a'** of first region **12a** of frame **12**. In some embodiments, the first surface **20a** of first magnet **20** will project for a distance outwardly beyond the front surface **12a'** of first region **12a** of frame **12** and may therefore project for a distance greater than "D" outwardly beyond outer surface **16a** of lens **16**. The distance "D" is about 1 mm in one embodiment of eyeglasses **10**. Sometimes, if lenses **16**, **18** are thicker than the frame **12**, the position of the outer surface **16a**, **18a** of the respective lens **16**, **18** is a more critical determinant of the overall length of the magnet **20**, **22**. In

order to keep the sunglass lenses away from the outer surface **16a**, **18a** of eyeglasses **10**, as will be described later herein, both the magnets **20**, **22** and metal rods associated with the sunglasses need to protrude outwardly from the associated lenses to ensure that the surfaces of the lenses of the eyeglasses **10** and those of the sunglasses do not contact one another. If one lens surface is touching another, the lenses may develop abrasions if dust or other particulates are present between the lens surfaces.

[0051] The thickness “T3” of the magnet **20**, **22** is greater than the thickness of the frame **12** may be longer than the thickness of the frame **12** such that the magnets **20**, **22** protrude past the outer surface **16a**, **18a** of the eyeglass lenses **16**, **18** by about 1 mm. Alternatively, the thickness of the magnet **20**, **22** should be such that the first end **20a**, **20a** of the magnet **20**, **22** is flush with the outermost surface **12a** of the frame **12**.

[0052] It will be understood that since first and second magnets **20**, **22** are cylindrical in shape and therefore have a circular circumferential exterior surface while notches **16d**, **18d** are U-shaped, the magnets **20**, **22** are not precisely complementary in shape to the notches **16d**, **18d**. Only the circumferential surface **20c**, **22c** of first and second magnets **20**, **22** with the same radius of curvature as the arcuate innermost edge (**16d'** for example) of the associate lens **16**, **18** is complementary to the notch **16d**, **18d**.

[0053] The rest of the wall of the first lens **16** and second lens **18** which define the respective notch **16d**, **18d** is not complementary to the rest of the circumferential surface **20c**, **22c** of magnets **20**, **22**. As a consequence, only portions of the first and second magnets **20**, **22** are in direct abutting contact with the wall which defines the notches **16d**, **18d** in lenses **16**, **18**.

[0054] Frame **12** holds lenses **16**, **18** in place. The interlocking of the frame **12** and lenses **16**, **18** holds the magnets **20**, **22** in place. There are many different metal or plastic frame designs that will hold a lens in place. Complete rimless designs of eyeglasses will not be able to be used in the performance of the disclosed method. Almost all frames have sufficient flexibility to accommodate and hold lenses **16**, **18** without damaging them. Lenses **16**, **18** have to be edged to the correct size so as not to be too loose within the frame **12** or they will fall out of the frame **12** and may break or be damaged. The lenses **16**, **18** cannot be too large for the frame as this may cause deformation of the frame and/or breakage of the frame and/or lenses **16**, **18**. With the lenses **16**, **18** interlocked within frame **12**, there is an equilibrium between the two. In order to install the magnets **20**, **22**, after the respective lens **16**, **18** has been edged to the correct size, the notch **16d**, **18d** is filed into or cut into the lens **16**, **18** to the appropriate depth. In a next step, the magnet **20**, **22** is inserted into the notch **16d**, **18d** and the lens **16** or **18** is then compressed into the frame **12** if the frame is an acetate frame. If the frame **12** is a metal frame, the eye wire will be held in place until the set screw is secured. The diameter of each magnet **20**, **22** is such that the magnets **20**, **22** may be retained partially within the associated notch **16d**, **18d** by friction exerted by the respective lens **16**, **18** and partially by a portion of the frame **12** which closes off access to the opening to the notch **16d**, **18d**. In other instances, the magnets **20**, **22** are entirely retained within notches **16d**, **18d** and thereby within lenses **16**, **18** by the frame **12** pinching or squeezing the lens **16**, **18**. In other words, the magnets **20**, **22** are held in place in the associated lens **16**, **18** by the pressure or pinch of the frame **12** and lens **16**, **18** together. This capturing of the magnet **20** within notch **16d** is illustrated in FIG. 4B. Small regions of the frame **12** which close off access to the opening to the notch **16d** may also assist in retaining the magnet **20** within the lens **16** and within notch **16d** thereof. The same arrangement is true of second magnet **22**, notch **18d**, lens **18** and frame **12**. Magnets **20**, **22** are essentially “pinched” or “squeezed” between the frame **12** and the associated lens **16**, **18**. At least part of the exterior circumferential surface **20c**, **22c** of each respective magnet **20**, **22** is not surrounded by material of the associated lens **16**, **18**.

[0055] Referring now to FIGS. 6 through 8B, there is shown the first embodiment eyeglasses **10** described above along with a first embodiment of sunglasses in accordance with an aspect of the present disclosure. The sunglasses are generally indicated by the reference number **30**. The

sunglasses **30**, as illustrated, comprise a single continuous sheet of lens material which is fabricated to form a first lens **32** and a second lens **34** of the sunglasses **30**. As disclosed earlier herein, secondary lenses may be sunglass lenses or any one of a range of therapeutic lenses for concussion, migraine etc. that are a specified tint not used for sunglasses. Others secondary lenses could include a filter for blue light emitted by monitors, clear, sunglass bifocals and readers for presbyopic people.

[0056] A bridge region **36** connects the first lens **32** and second lens **34** to one another. The sunglasses **30** have an outer surface **30a** and an inner surface **30b** and a perimeter edge **30c** which extends between the outer surface **30a** and inner surface **30b**. The overall configuration of the first lens **32** and second lens **34** is generally similar to the overall exterior shape of frame **12** with first lens **16** and second lens **18** therein. Sunglasses **30** are configured to be operatively engaged with eyeglasses **10** as indicated by arrow “E” in FIG. **6**. The manner of operatively engaging sunglasses **30** with eyeglasses **10** will be described later herein.

[0057] In accordance with an aspect of the present disclosure, a first aperture **32a** is defined in first lens **32** and a second aperture **34a** is defined in second lens. The apertures **32a**, **34a** extend between the outer surface **30a** of the associated lens **32**, **34** and the inner surface **30b** thereof. Each aperture **32a**, **32b** is located a distance inwardly away from the perimeter edge **30c** of sunglasses **30**. In particular, apertures **32a**, **32b** are located so as to selectively alignable with the magnets **20**, **22** of eyeglasses **10** when sunglasses **30** are operatively engaged with eyeglasses **10**.

[0058] In accordance with a further aspect of the present disclosure, a first rod **36** and a second rod **38** are operatively engaged in first aperture **32a** and second aperture **34a** respectively. Rods **36**, **38** are fabricated from metals or metal alloys and can be of any suitable cross section and shape. Even custom shapes can be fabricated to provide a more aesthetically appealing and interesting appearance to sunglasses **30**. Each of the first rod **36** and second rod **38** is tightly retained within the associated first aperture **32a** and second aperture **34a** in any suitable manner. Suitable manners of retaining the rods **36**, **38** in place include but are not limited to press-fitting the rods **36**, **38** into the sunglasses' lens material and retaining the same therein through friction. Adhesives are not utilized to hold the rods **36**, **38** in place. It will be understood, however, that in other embodiments, adhesives or other fastening mechanisms may be utilized instead of press-fitting.

[0059] As best seen in FIG. **8A**, first rod **36** includes a first end **36a**, a second end **36b** opposed to first end **36a**, and a perimeter surface **36c** extending between first end **36a** and second end **36b**. The thickness of first rod **36** is indicated as thickness “T4” as measured between first end **36a** and second end **36b**. The thickness of the sunglasses' lens material is indicated as thickness “T5”. As is evident from FIG. **8A**, thickness “T4” of first rod **36** is greater than thickness “T5” of the sunglasses' lens material. First rod **36** is arranged in the lens material of sunglasses **30** in such a way that a portion of the first rod **36** extends outwardly for a distance beyond the inner surface **30b** of the lens **32**. First rod **36** is arranged such that first end **36a** thereof is substantially flush with outer surface **30a** of the lens **32** of sunglasses **30**. It will be understood that second rod **38** is similarly arranged in sunglasses **30**.

[0060] In the first embodiment sunglasses **30**, the configuration of first rod **36** and second rod **38** is substantially similar to the configuration of the magnets **20**, **22** in eyeglasses **10**. As illustrated, each of the magnets **20**, **22** is generally cylindrical in configuration and presents a generally circular first end **20a**, **22a** which is located a distance outwardly beyond an outer surface of the associated first lens **16** and second lens **18**, respectively, of eyeglasses **10**. Similarly, each of the first and second rods **36**, **38** is generally cylindrical and presents a generally circular second end (such as second end **36b**) which is located a distance outwardly beyond an inner surface **30b** of sunglasses **30**. As illustrated in FIGS. **8A** and **8B**, in some embodiments, the diameter of the first and second magnets **20**, **22** and the diameter of the first and second rods **36**, **38** is substantially the same. It will be understood, however, that in other embodiments, the diameter of the first and second magnets **20**, **22** may be greater than or smaller than the diameter of the first and second rods

36, 38.

[0061] In accordance with an aspect of the present disclosure, each of the first rod **36** and the second rod **38** is comprised of a non-magnetic material. In particular, each of the first rod **36** and second rod **38** is comprised of a metal and, more particularly, from a non-magnetic metal. In other words, the metal first and second rods **36, 38** utilized in sunglasses **30** do not have a north pole or a south pole, nor do they exhibit a magnetic field in of themselves. One suitable metal for first rod **36** and second rod **38** is steel. In one embodiment, the steel is stainless steel. Specifically, the stainless steel is of a type that is capable of being attracted to a magnet. The metal used for first rod **36** and second rod **38** is not, in of itself, magnetic. In other words, the metal is incapable of magnetically attracting other metals towards itself. The metal used in rods **36, 38** may be of a type, however, that is able to be attracted to a magnet.

[0062] Sunglasses **30** are operatively engaged with eyeglasses **10** by moving sunglasses **30** towards eyeglasses **10** or vice versa as indicated by arrow “E” in FIG. **6**. The movement “E” towards one another is continued until first and second rods **36, 38** are aligned with first and second magnets **20, 22** and are brought into abutting physical contact with one another. When first and second rods **36, 38** are either close enough to first and second magnets **20, 22** or are in physical contact with the same, the magnetic fields of first and second magnets **20, 22** attracts the non-magnetic material of first and second rods **36, 38** thereto. T Sunglasses **30** are thus retained in operative engagement with eyeglasses **10**.

[0063] As is evident from FIG. **8B**, when sunglasses **30** are operatively engaged with eyeglasses **10** via magnets **20, 22** and rods **36, 38**, a space **40** is defined between the outer surface **16a** of first lens **16** (and the outer surface **18a** of second lens **18**) and the inner surface **30b** of sunglasses **30**. The space **40** is of a size that is of a size equal to a thickness of the portions of magnets **20, 22** which extend beyond the outer surfaces **16a, 18a** of the associated lenses **16, 18** plus the thickness of the portions of the rods **36, 38** which extend beyond the inner surface **30b** of sunglasses **30**. When it is desired to remove sunglasses **30** from eyeglasses **10**, sunglasses **30** are simply pulled away from eyeglasses **10** and the attraction between first and second magnets **20, 22** and first and second rods **36, 38** is broken. The space **40** ensures that the sunglasses **30** are not brought into physical contact with the outer surfaces **16a, 18a** of the lenses **16, 18** and thus reduces the possibility that the sunglasses **30** might inadvertently scratch or otherwise damage the lenses **16, 18**.

[0064] FIG. **9** shows a second embodiment of sunglasses **130** which are operatively engageable, as indicated by arrow “E”, with eyeglasses **10**. Sunglasses **130** differ in configuration from sunglasses **30** in that the sunglasses **130** comprise two separate lenses **132, 134** which are connected to one another by a bridge **136**. It will be understood that lenses **132, 134** may be prescription lenses. Each of the first lens **132** and second lens **134** is provided with a first rod **136** and a second rod **138**, respectively. First rod **136** and second rod **138** are substantially identical in all respects to first rod **36** and second rod **38** except for the overall shape and size thereof. While first rod **36** and second rod **38** are cylindrical in configuration and present a circular second end for physical contact with the circular first ends **20a, 22a** of magnets **20, 22**, first and second rods **136, 138** are cuboid in configuration and present a square second end for physical contact with the circular first ends **20a, 22a** of magnets **20, 22**. Otherwise, sunglasses **130** are operatively engageable and disengageable with eyeglasses **10** in substantially an identical manner to sunglasses **30**. It will be understood that the configuration of magnets **20, 22** and rods **136, 138** may be reversed, i.e., magnets **20, 22** may be cuboid in configuration while rods **136, 138** may be cylindrical in configuration. It will further be understood that magnets **20, 22**, and rods **136, 138** (or **36, 38**) may be of any desired configurations, and may be similar or dissimilar to one another. All that is needful is that the magnets **20, 22** and rods **136, 138** (or **36, 38**) present a sufficient degree of alignable surface area to one another to enable engagement of sunglasses **130** (or **30**) with eyeglasses **10**.

[0065] FIG. **10** illustrates the fact instead of the magnets being provided in the eyeglasses and the rods being provided in the sunglasses, the materials may be reversed. In other words, the rods (non-

magnetic material) may be provided in the eyeglasses and the magnets may be provided in sunglasses. FIG. 10 shows a portion of a lens 416 of eyeglasses 410 which has a non-magnetic rod 436 therein and a portion of a lens 432 of sunglasses 430 which has a magnet 420 therein. The sunglasses 430 are retained in engagement with the eyeglasses 410 by the attraction between the magnet 420 and stainless steel rod 436. The attraction is indicated in FIG. 8A by the arrows "F". [0066] Referring now to FIGS. 11-11B, a further embodiment of eyeglasses is illustrated, generally indicated at 510. Eyeglasses 510 as illustrated include what is known as a NYLOR® frame. (NYLOR® is a registered trademark of Nylor of Gif-sur-yvette, FRANCE). In this type of eyeglasses, a full frame, such as frame 12 of eyeglasses 10, is replaced with a frame member 542 and a nylon line 544. Together, the frame member 542 and nylon line 544 bound and define two apertures 546 and 548 into which are received a first lens 516 and a second lens 518. Each of the lenses 516 and 518 of eyeglasses 510 are substantially identical to lenses 16, 18 except for the characteristics described hereafter. Each lens 516, 518 defines a notch 516d, 518d into which a magnet 520, 522 (or first or second rod), respectively, is received. As is illustrated in FIGS. 11A and 11B, the first lens 516 (and similarly the second lens 518) is configured to define a channel 516e in the peripheral edge 516c thereof. The channel 516e extends at least around a first side, a second side, and a bottom end of the first lens 516. The channel 516e may additionally extend along a top end of the first lens 516 but in other embodiments the channel may not be defined in the top end of the first lens 516 and the top end will simply be secured to the frame member 542 in any known manner. A similar arrangement exists with second lens 518. Channel 516e is specifically configured to be complementary to at least a portion of the circumferential surface of the nylon line 544. FIGS. 11A and 11B show the nylon line 544 received in channel 516e of first lens 516. In this particular style of eyeglasses 510, the magnets 520, 522 (or rods) are retained within lenses 516, 518 by a combination of friction and the nylon line 544 pinching or squeezing the magnets 520, 522 inwardly in contact with the lens material defining notches 516d, 518d. Eyeglasses 510 are used in the same manner as eyeglasses 10 and all aspects of the structure and function of eyeglasses 510 and sunglasses 30, 130, as described earlier herein, are applicable to eyeglasses 510.

[0067] It will be understood that the components utilized in the presently disclosed method may be off-the-shelf products. The components do not need to be specially manufactured in order to work the present disclosure. The use of off-the-shelf products helps reduce the cost involved to enable sunglasses to be engaged with eyeglasses.

[0068] Furthermore, the present methodology can be used to retrofit an existing or new sunglasses to a new or existing eyeglasses. Alternatively, the methodology may be performed on a newly ordered eyeglasses and/or sunglasses. The present methodology may furthermore be performed without a specific software program on regular lens edging equipment or milling machines and requires minimal tools. For example, a punch is able to be utilized to form the notches in the eyeglass lenses. Because the notches 16d, 18d, originate in the peripheral edge 16c, 18c of the associated lens 16, 18 and extend inwardly therefrom for a distance, the positioning of the one of the magnetic member and the non-magnetic member in the eyeglass lenses is less obtrusive than it otherwise could be. Because the one of the magnetic member and the non-magnetic member is adjacent the peripheral edge 16c, 18c of the associated lens there is little to no impact on a person's vision through the lenses.

[0069] It will further be understood that if a person decides to purchase a new eyeglasses and/or a new sunglasses, the magnetic members and/or non-magnetic members may be removed from an old eyeglasses or sunglasses and may be readily installed in the new glasses. In other words, the magnetic members and/or non-magnetic members may be reused.

[0070] Various inventive concepts may be embodied as one or more methods, of which an example has been provided. The acts performed as part of the method may be ordered in any suitable way. Accordingly, embodiments may be constructed in which acts are performed in an order different than illustrated, which may include performing some acts simultaneously, even though shown as

sequential acts in illustrative embodiments.

[0071] While various inventive embodiments have been described and illustrated herein, those of ordinary skill in the art will readily envision a variety of other means and/or structures for performing the function and/or obtaining the results and/or one or more of the advantages described herein, and each of such variations and/or modifications is deemed to be within the scope of the inventive embodiments described herein. More generally, those skilled in the art will readily appreciate that all parameters, dimensions, materials, and configurations described herein are meant to be exemplary and that the actual parameters, dimensions, materials, and/or configurations will depend upon the specific application or applications for which the inventive teachings is/are used. Those skilled in the art will recognize, or be able to ascertain using no more than routine experimentation, many equivalents to the specific inventive embodiments described herein. It is, therefore, to be understood that the foregoing embodiments are presented by way of example only and that, within the scope of the appended claims and equivalents thereto, inventive embodiments may be practiced otherwise than as specifically described and claimed. Inventive embodiments of the present disclosure are directed to each individual feature, system, article, material, kit, and/or method described herein. In addition, any combination of two or more such features, systems, articles, materials, kits, and/or methods, if such features, systems, articles, materials, kits, and/or methods are not mutually inconsistent, is included within the inventive scope of the present disclosure.

[0072] The articles “a” and “an,” as used herein in the specification and in the claims, unless clearly indicated to the contrary, should be understood to mean “at least one.” The phrase “and/or,” as used herein in the specification and in the claims (if at all), should be understood to mean “either or both” of the elements so conjoined, i.e., elements that are conjunctively present in some cases and disjunctively present in other cases. Multiple elements listed with “and/or” should be construed in the same fashion, i.e., “one or more” of the elements so conjoined. Other elements may optionally be present other than the elements specifically identified by the “and/or” clause, whether related or unrelated to those elements specifically identified. Thus, as a non-limiting example, a reference to “A and/or B”, when used in conjunction with open-ended language such as “comprising” can refer, in one embodiment, to A only (optionally including elements other than B); in another embodiment, to B only (optionally including elements other than A); in yet another embodiment, to both A and B (optionally including other elements); etc. As used herein in the specification and in the claims, “or” should be understood to have the same meaning as “and/or” as defined above. For example, when separating items in a list, “or” or “and/or” shall be interpreted as being inclusive, i.e., the inclusion of at least one, but also including more than one, of a number or list of elements, and, optionally, additional unlisted items. Only terms clearly indicated to the contrary, such as “only one of” or “exactly one of,” or, when used in the claims, “consisting of,” will refer to the inclusion of exactly one element of a number or list of elements. In general, the term “or” as used herein shall only be interpreted as indicating exclusive alternatives (i.e. “one or the other but not both”) when preceded by terms of exclusivity, such as “either,” “one of,” “only one of,” or “exactly one of.” “Consisting essentially of,” when used in the claims, shall have its ordinary meaning as used in the field of patent law.

[0073] As used herein in the specification and in the claims, the phrase “at least one,” in reference to a list of one or more elements, should be understood to mean at least one element selected from any one or more of the elements in the list of elements, but not necessarily including at least one of each and every element specifically listed within the list of elements and not excluding any combinations of elements in the list of elements. This definition also allows that elements may optionally be present other than the elements specifically identified within the list of elements to which the phrase “at least one” refers, whether related or unrelated to those elements specifically identified. Thus, as a non-limiting example, “at least one of A and B” (or, equivalently, “at least one of A or B,” or, equivalently “at least one of A and/or B”) can refer, in one embodiment, to at least

one, optionally including more than one, A, with no B present (and optionally including elements other than B); in another embodiment, to at least one, optionally including more than one, B, with no A present (and optionally including elements other than A); in yet another embodiment, to at least one, optionally including more than one, A, and at least one, optionally including more than one, B (and optionally including other elements); etc.

[0074] While components of the present disclosure are described herein in relation to each other, it is possible for one of the components disclosed herein to include inventive subject matter, if claimed alone or used alone. In keeping with the above example, if the disclosed embodiments teach the features of A and B, then there may be inventive subject matter in the combination of A and B, A alone, or B alone, unless otherwise stated herein.

[0075] As used herein in the specification and in the claims, the term “effecting” or a phrase or claim element beginning with the term “effecting” should be understood to mean to cause something to happen or to bring something about. For example, effecting an event to occur may be caused by actions of a first party even though a second party actually performed the event or had the event occur to the second party. Stated otherwise, effecting refers to one party giving another party the tools, objects, or resources to cause an event to occur. Thus, in this example a claim element of “effecting an event to occur” would mean that a first party is giving a second party the tools or resources needed for the second party to perform the event, however the affirmative single action is the responsibility of the first party to provide the tools or resources to cause said event to occur.

[0076] When a feature or element is herein referred to as being “on” another feature or element, it can be directly on the other feature or element or intervening features and/or elements may also be present. In contrast, when a feature or element is referred to as being “directly on” another feature or element, there are no intervening features or elements present. It will also be understood that, when a feature or element is referred to as being “connected”, “attached” or “coupled” to another feature or element, it can be directly connected, attached or coupled to the other feature or element or intervening features or elements may be present. In contrast, when a feature or element is referred to as being “directly connected”, “directly attached” or “directly coupled” to another feature or element, there are no intervening features or elements present. Although described or shown with respect to one embodiment, the features and elements so described or shown can apply to other embodiments. It will also be appreciated by those of skill in the art that references to a structure or feature that is disposed “adjacent” another feature may have portions that overlap or underlie the adjacent feature.

[0077] Spatially relative terms, such as “under”, “below”, “lower”, “over”, “upper”, “above”, “behind”, “in front of”, and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. It will be understood that the spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. For example, if a device in the figures is inverted, elements described as “under” or “beneath” other elements or features would then be oriented “over” the other elements or features. Thus, the exemplary term “under” can encompass both an orientation of over and under. The device may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein interpreted accordingly. Similarly, the terms “upwardly”, “downwardly”, “vertical”, “horizontal”, “lateral”, “transverse”, “longitudinal”, and the like are used herein for the purpose of explanation only unless specifically indicated otherwise.

[0078] Although the terms “first” and “second” may be used herein to describe various features/elements, these features/elements should not be limited by these terms, unless the context indicates otherwise. These terms may be used to distinguish one feature/element from another feature/element. Thus, a first feature/element discussed herein could be termed a second feature/element, and similarly, a second feature/element discussed herein could be termed a first

feature/element without departing from the teachings of the present invention.

[0079] An embodiment is an implementation or example of the present disclosure. Reference in the specification to “an embodiment,” “one embodiment,” “some embodiments,” “one particular embodiment,” “an exemplary embodiment,” or “other embodiments,” or the like, means that a particular feature, structure, or characteristic described in connection with the embodiments is included in at least some embodiments, but not necessarily all embodiments, of the invention. The various appearances “an embodiment,” “one embodiment,” “some embodiments,” “one particular embodiment,” “an exemplary embodiment,” or “other embodiments,” or the like, are not necessarily all referring to the same embodiments.

[0080] If this specification states a component, feature, structure, or characteristic “may”, “might”, or “could” be included, that particular component, feature, structure, or characteristic is not required to be included. If the specification or claim refers to “a” or “an” element, that does not mean there is only one of the element. If the specification or claims refer to “an additional” element, that does not preclude there being more than one of the additional element.

[0081] As used herein in the specification and claims, including as used in the examples and unless otherwise expressly specified, all numbers may be read as if prefaced by the word “about” or “approximately,” even if the term does not expressly appear. The phrase “about” or “approximately” may be used when describing magnitude and/or position to indicate that the value and/or position described is within a reasonable expected range of values and/or positions. For example, a numeric value may have a value that is $\pm 0.1\%$ of the stated value (or range of values), $\pm 1\%$ of the stated value (or range of values), $\pm 2\%$ of the stated value (or range of values), $\pm 5\%$ of the stated value (or range of values), $\pm 10\%$ of the stated value (or range of values), etc. Any numerical range recited herein is intended to include all sub-ranges subsumed therein.

[0082] Additionally, the method of performing the present disclosure may occur in a sequence different than those described herein. Accordingly, no sequence of the method should be read as a limitation unless explicitly stated. It is recognizable that performing some of the steps of the method in a different order could achieve a similar result.

[0083] In the claims, as well as in the specification above, all transitional phrases such as “comprising,” “including,” “carrying,” “having,” “containing,” “involving,” “holding,” “composed of,” and the like are to be understood to be open-ended, i.e., to mean including but not limited to. Only the transitional phrases “consisting of” and “consisting essentially of” shall be closed or semi-closed transitional phrases, respectively.

[0084] To the extent that the present disclosure has utilized the term “invention” in various titles or sections of this specification, this term was included as required by the formatting requirements of word document submissions pursuant the guidelines/requirements of the United States Patent and Trademark Office and shall not, in any manner, be considered a disavowal of any subject matter.

[0085] In the foregoing description, certain terms have been used for brevity, clearness, and understanding. No unnecessary limitations are to be implied therefrom beyond the requirement of the prior art because such terms are used for descriptive purposes and are intended to be broadly construed.

[0086] Moreover, the description and illustration of various embodiments of the disclosure are examples and the disclosure is not limited to the exact details shown or described.

Claims

1. An eyewear assembly comprising: primary lenses; secondary lenses; a connector assembly detachably securing the primary lenses and the secondary lenses to one another; wherein the connector assembly comprises: a magnetic member provided in one of the primary lenses and the secondary lenses; and a non-magnetic member provided in the other of the primary lenses and the secondary lenses.

2. The eyewear assembly according to claim 1, wherein the non-magnetic member is comprised of a material which is attractable to the magnetic member.
3. The eyewear assembly according to claim 2, wherein the material is a metal.
4. The eyewear assembly according to claim 3, wherein the metal is stainless steel.
5. The eyewear assembly according to claim 1, wherein a region of the magnetic member extends outwardly beyond a surface of an associated lens of the one of the primary lenses and the secondary lenses.
6. The eyewear assembly according to claim 1, wherein a region of the non-magnetic member extends outwardly beyond a surface of an associated lens of the other of the primary lenses and the secondary lenses.
7. The eyewear assembly according to claim 1, wherein, when the secondary lenses are operatively engaged with the primary lenses, an end surface of the magnetic member in the one of the primary lenses and the secondary lenses contacts an end surface of the non-magnetic member in the other of the primary lenses and the secondary lenses.
8. The eyewear assembly according to claim 7, further comprising a space defined between an outer surface of the lens of the primary lenses and an inner surface of the lens of the secondary lenses.
9. The eyewear assembly according to claim 1, wherein a notch originates in a peripheral edge of the lens of the primary lenses and the one of the magnetic member and the non-magnetic member is received in the notch.
10. The eyewear assembly according to claim 9, wherein the notch extends between an outer surface and an inner surface of the lens of the primary lenses.
11. The eyewear assembly according to claim 9, wherein a region of a wall of the lens which defines the notch is complementary in configuration to a portion of a circumferential surface of the one of the magnetic member and the non-magnetic member received in the notch.
12. The eyewear assembly according to claim 9, further comprising a securement mechanism provided on the primary lenses, wherein the securement mechanism retains the one of the magnetic member and the non-magnetic member in the notch.
13. The eyewear assembly according to claim 12, wherein the securement mechanism comprises a portion of a frame of the primary lenses.
14. The eyewear assembly according to claim 12, wherein the securement mechanism comprises a length of a nylon wire provided on the primary lenses.
15. The eyewear assembly according to claim 1, wherein an aperture extends between an outer surface and an inner surface of the lens of the secondary lenses, wherein the aperture is located a distance inwardly away from a peripheral edge of the lens of the secondary lenses, and wherein the other of the magnetic member and the non-magnetic member is received in the aperture.
16. A method comprising: providing primary lenses including at least one eyeglass lens; providing secondary lenses including at least one sunglass lens; providing a magnetic member; providing a non-magnetic member; installing one of the magnetic member and the non-magnetic member on the at least one eyeglass lens; installing the other of the magnetic member and the non-magnetic member on the at least one sunglass lens; attracting the non-magnetic member to the magnetic member; and retaining the secondary lenses to the primary lenses.
17. The method according to claim 16, further comprising: defining a notch in the at least one eyeglass lens; defining an aperture in the at least one sunglass lens; installing the one of the magnetic member and the non-magnetic member in the notch defined in the at least one eyeglass lens; and installing the other of the magnetic member and the non-magnetic member in the aperture defined in the at least one sunglass lens.
18. The method according to claim 16, further comprising: moving the secondary lenses and primary lenses into close proximity with one another; and aligning the magnetic member in the one of the primary lenses and the secondary lenses with the non-magnetic member in the other of the

primary lenses and the secondary lenses.

19. The method according to claim 18, further comprising bringing the magnetic member and the non-magnetic member into direct contact with one another.

20. The method according to claim 16, wherein providing the non-magnetic member includes providing a rod of stainless steel capable of being attracted to a magnet.
